

Dataset Overview & Project Introduction

Continuous Controls Monitoring, Procure to Pay | Vendor Invoice Ledger | 12,095 Invoices | FY 2025-26

Source & Nature

Synthetic vendor invoice ledger built for academic use, with control exceptions seeded for each test. 12,095 invoices, 11 variables, April 2025 to March 2026.

Scope & Coverage

- ₹123.8 Cr total spend
- 150 vendors across 8 spend categories
- 8 invoice creators, 5 approvers
- Posting date and time for every invoice

Key Variables

Invoice ID · Vendor · Category · Invoice No.
Invoice Date · Posted At · Amount
PO Number · PO Amount · Created By · Approved By

Business Relevance

- Accounts payable is the largest routine cash outflow
- Tests whether P2P controls stop duplicate, unauthorised or unsupported payments
- Replaces sample testing with full population coverage

Control Tests

Six procure to pay controls, each tested on the full population of 12,095 invoices

1 Duplicate payments

Has the same invoice been paid more than once?

Rule: Same vendor and amount within 30 days, or a reused invoice number

2 Approval threshold avoidance

Are invoices sized or split to avoid the second approver?

Rule: ₹4.5L to ₹5L, or same day splits that together reach ₹5L

3 Benford's Law, by vendor

Do a vendor's amounts behave like genuine transactions?

Rule: Chi-square on first digits, $p = 0.001$, vendors with 30+ invoices

4 Three way match

Are we paying more than was ordered, or buying without a PO?

Rule: Invoice above PO by more than 5%, or no PO at ₹1L and above

5 Segregation of duties

Can one person raise and approve the same invoice?

Rule: Invoice creator is also the approver

6 After hours posting

Are entries posted when no one is reviewing?

Rule: Posted on a Sunday, or between 22:00 and 06:00

Problem Statement

Core Problem

Accounts payable controls are commonly tested on a small sample of invoices. A sample can show that a control exists, but not where it fails: a duplicate, a split invoice or a self approved payment in the untested population goes unnoticed, and each is cash that has already left the business. Without full population testing, management cannot see how often controls fail, which vendors and users are involved, or how much money is exposed.

Objective

Test every invoice in the year against six P2P controls, quantify the exceptions by count and value, and identify where control failures concentrate by vendor, user and period.

Research Questions

- Which control fails most often, and which exposes the most value?
- Are exceptions concentrated in a few vendors or users?
- Which failures exceed performance materiality and need priority action?

Expected Outcome

A repeatable monitoring routine with a dashboard that drills from test summary to transaction, and a prioritised set of preventive control changes for management.

Methodology

Four step analytical process in Python and Power BI Desktop

Data Preparation

Imported CSV into pandas → parsed dates and timestamps → checked nulls, data types and duplicate invoice IDs → normalised invoice numbers → 12,095 clean rows

Control Test Design

Translated each control into a rule with an adjustable parameter → coded six tests in Python → one exception row per invoice per failed test, with the reason

Exception Analysis

Counted unique exceptions and value flagged → compared each test with performance materiality of ₹92.9 lakh → identified vendor and user concentration

Visualisation & Reporting

Built a 4 page Power BI dashboard (summary, Benford, vendor risk, transaction drill-through) → wrote a findings memo in condition, criteria, cause, effect and action form

Tools: Python (pandas, NumPy) Power BI Desktop (DAX, drill-through) Excel (validation)

Key Insights

Five findings from full population testing of the FY 2025-26 ledger

12,095

Invoices tested, ₹123.8 Cr

1,152

Unique exceptions, 9.5%

₹25.5 Cr

Value flagged, 20.6% of spend

4 of 6

Tests above performance materiality

Threshold avoidance is the largest exposure

315 invoices worth ₹12.9 Cr: 193 sit just under the ₹5L approval limit and 122 are same day splits. Approval routing checks single invoices, not what a vendor bills in a day.

Three way match failures follow closely

298 invoices worth ₹10.6 Cr: 145 exceed their PO by more than 5% and 153 are non PO invoices of ₹1L and above.

Benford flags six vendors, three strongly

Apex Logistics, Summit Media and Zenith Printers have no amounts starting with 1 to 4, a typical sign of invented invoices. The other three are borderline and may be false positives.

Duplicates are direct cash leakage

189 invoices worth ₹1.9 Cr repeat an earlier invoice. Any that were paid are recoverable losses.

Access and timing weaknesses

94 invoices (₹82 lakh) were approved by their own creator, across all eight creators, so the workflow itself allows it. 86 were posted after hours (₹50 lakh), 36 of them on Sundays.

Business Recommendations

Four control actions, prioritised by value exposed

Preventive ERP configuration

Block duplicate invoices on vendor, amount and normalised invoice number. Put a hard stop on invoices above PO tolerance and make a PO mandatory at ₹1L and above. Raise recovery claims for duplicates already paid.

Approval workflow redesign

Route second approval on cumulative vendor spend per day, not per invoice. Review the purchasing pattern of vendors whose invoices cluster just below the ₹5L limit.

Vendor and access review

Obtain supporting documents and confirm receipt of goods or services for the three high risk Benford vendors. Remove creator and approver access conflicts and add a system rule that blocks self approval.

Continuous monitoring cadence

Run the six tests monthly on the full population. Route every exception to an owner, record a disposition (cleared or escalated) and track closure on the dashboard.

Dashboard — Exception Overview

Dashboard view · Page 1 of 3

CONTINUOUS CONTROLS MONITORING · PROCURE TO PAY · FY 2025-26

Invoices tested

12,095

full population

Value tested

₹123.8 Cr

FY 2025-26

Exceptions

1,152

unique invoices

Value flagged

₹25.5 Cr

unique invoices

Exception rate

9.5%

of invoices

Value at risk

20.6%

of spend

Vendors flagged

138

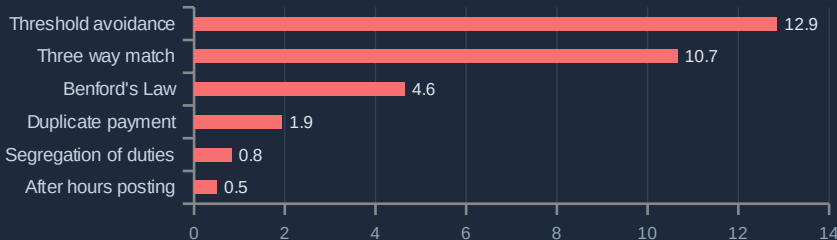
of 150 vendors

Tests above PM

4 of 6

PM ₹92.9 lakh

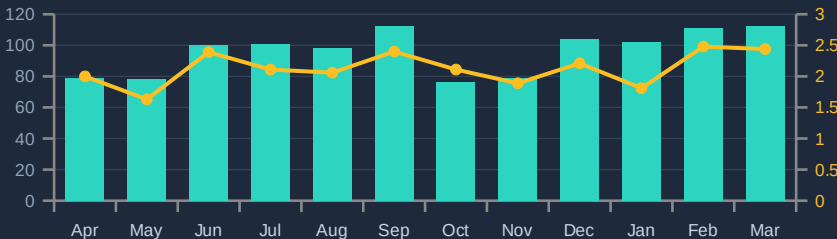
Value flagged by test, ₹ Cr



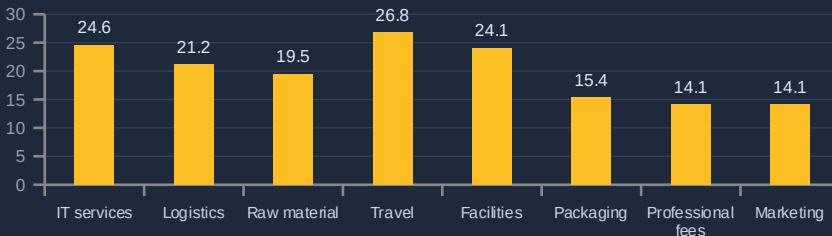
Exceptions by test, count



Monthly trend: exceptions (bars) and value flagged, ₹ Cr (line)



Value at risk by spend category, % of category spend



Two tests carry most of the value · Exceptions recur every month · Travel and IT services have the highest share of spend flagged

Dashboard — Test Analytics

Dashboard view · Page 2 of 3

CONTINUOUS CONTROLS MONITORING · PROCURE TO PAY · FY 2025-26

Duplicate Invoices

189

₹1.94 Cr to recover

Just under limit

193

₹4.5L to ₹5L

Same day splits

122

invoices

Over PO by >5%

145

invoices

Non PO ≥ ₹1L

153

invoices

Benford failures

6 vendors

3 strong signals

Self approved

94

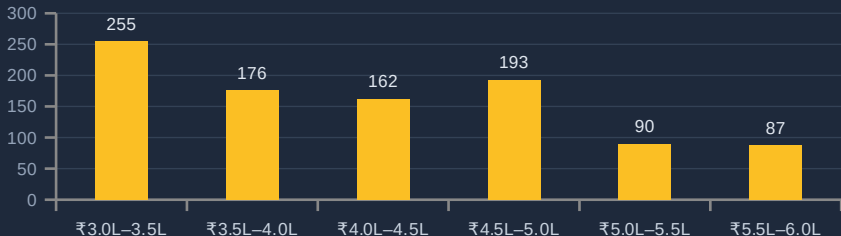
all 8 creators

After hours

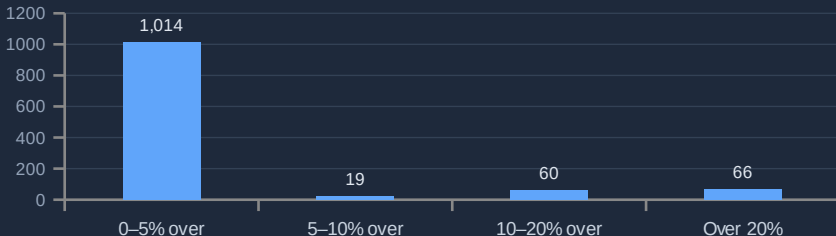
86

36 on Sundays

Invoices by amount band around the ₹5L approval limit



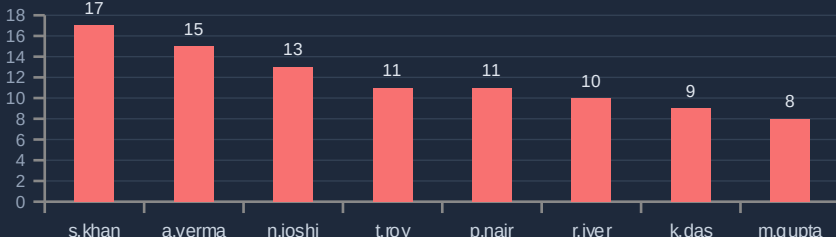
Invoices above PO value (9,156 at or below PO not shown)



Benford first digit test, % of invoices



Self approved invoices by user (segregation of duties)



Invoice counts rise just below ₹5L and halve above it · 1,014 invoices sit inside the 5% PO tolerance · Self approval occurs for every creator, so it is a configuration gap

Dashboard — Vendor & User Risk

Dashboard view · Page 3 of 3

CONTINUOUS CONTROLS MONITORING · PROCURE TO PAY · FY 2025-26

Vendors tested

150

all active vendors

Vendors with exceptions

138

at least one

Top 10 vendor share

49.9%

of value flagged

Highest risk vendor

Vertex Travels

₹2.18 Cr flagged

Strong Benford signals

3 vendors

χ^2 above 130

Creator exception rate

8.9 to 10.2%

range across 8 users

Vendor risk heatmap: exceptions per test, top 10 vendors by value flagged

Vendor	Dup	Limit	Benford	PO	SoD	Hours	₹ Cr
Vertex Travels	6	28	80	3	1		2.18
Bharat Electricals	7	17		19	1	2	2.00
Sai Media	3	23		14	2	4	1.67
Blue Ocean Solutions	3	25		11	1	1	1.18
Zenith Solutions	3	20	25	3	1	1	1.14
Apex Printers	1	7		11	2		0.98
Sunrise Media	6	20		9		2	0.93
Vardhman Printers	2	6		11	2	1	0.91
Pioneer Enterprises	1	20	22	20		1	0.89
Sai Electricals	1	8		8	2		0.86

Benford's Law: vendors failing the chi-square test

Vendor	Invoices	χ^2	Signal
Apex Logistics	64	163.9	Strong
Zenith Printers	64	152.9	Strong
Summit Media	72	137.9	Strong
Pioneer Enterprises	63	33.1	Borderline
Zenith Solutions	107	27.1	Borderline
Vertex Travels	229	26.4	Borderline

Invoice creators: exceptions and self approvals

User	Invoices	Exceptions	Rate	Self appr.
r.iyer	1,556	159	10.2%	10
n.joshi	1,546	154	10.0%	13
s.khan	1,491	149	10.0%	17
a.verma	1,509	148	9.8%	15
m.gupta	1,467	135	9.2%	8
k.das	1,454	132	9.1%	9
t.roy	1,538	139	9.0%	11
p.nair	1,534	136	8.9%	11

Ten vendors carry half the flagged value · Three vendors show strong Benford signals · Exception rates are even across users, pointing to process rather than individual failure

Key Learnings & AI Usage

Key Learnings

Full population testing changes the question

A sample shows whether a control works. The full population shows where it fails, how often and how much money is involved.

Parameters are judgement calls

Tightening PO tolerance from 5% to 2% raises exceptions from 1,152 to 1,610. Every threshold needs a documented rationale.

An exception is a lead, not a finding

Borderline Benford vendors show why each exception needs a review and a recorded disposition before it becomes a finding.

Count and value tell different stories

Benford produced the most exceptions (319), but threshold avoidance carried the most value (₹12.9 Cr).

AI Declaration

1. Dataset

The synthetic ledger was generated with AI assistance, with known exceptions seeded so each test has cases to find.

2. Code support

AI assisted in structuring the Python test logic. All tests were run, reviewed and reconciled by me.

3. Presentation

AI helped structure this deck and the findings memo format.

Appendix

Test results, dataset assumptions and variable definitions

Table 1 Control test results

Test	Rule	Exceptions	Value, ₹ Cr	vs PM
Threshold avoidance	₹4.5L to <₹5L, or same day splits ≥ ₹5L	315	12.85	Above
Three way match	> PO + 5%, or no PO at ≥ ₹1L	298	10.65	Above
Benford's Law	Vendor $\chi^2 > 26.12$, $n \geq 30$	319	4.64	Above
Duplicate payment	Vendor + amount ≤ 30 days, or reused number	189	1.94	Above
Segregation of duties	Creator = approver	94	0.82	Below
After hours posting	Sunday, or 22:00 to 06:00	86	0.50	Below
Unique exceptions	An invoice is counted once	1,152	25.53	

Data Sources & Assumptions

Dataset: demo_vendor_ledger_FY26.csv
Source: Synthetic AP ledger (academic use)
Period: April 2025 to March 2026
Overall materiality: 1% of spend (₹1.24 Cr)
Performance materiality: 75% of overall (₹92.9 lakh)

Variable Glossary

Amount / PO Amount: invoice value and ordered value
Created By / Approved By: user who raised and approved
Posted At: date and time of ERP posting
Exception: an invoice failing one or more tests
Performance materiality: threshold for priority findings